

MDSA D212 Data Mining II

Task 1 Clustering Techniques

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WGU

Scenario 1

One of the most critical factors in customer relationship management that directly affects a company's long-term profitability is understanding the customers. When a company understands its customers' characteristics, it is better able to target products and marketing campaigns for customers, resulting in better profits for the company in the long term.

You are an analyst for a telecommunications company that wants to better understand the characteristics of its customers. You have been asked to use clustering techniques to analyze customer data to identify groups of customers with similar characteristics, ultimately enabling better business and strategic decision-making.

Part I: Research Question

A. Describe the purpose of this data mining by doing the following:

1. Propose one question relevant to a real-world organizational situation that you will answer using one of the following clustering techniques:
 - k-means, using only continuous variables
 - hierarchical
 - Can customers be grouped into categories based on the survey responses available within the churn dataset using K-Means clustering to create better marketing? I believe this is an important question to an organization as it costs 10 times more to acquire a new customer than to retain an existing one(Churn Data Consideration and Dictionary 2024).

2. Define one goal of the data analysis. Ensure that your goal is reasonable within the scope of the scenario and is represented in the available data.
 - The goal of this analysis is to use K-Means clustering to define groups based off of the chosen variables and identify any patterns or correlation to help marketing with customer retention.

Part II: Method Justification

B. Explain the reasons for your chosen clustering technique from part A1 by doing the following::

1. Explain how the clustering technique you chose analyzes the selected data set. Include expected outcomes.
 - K-means clustering is an unsupervised learning algorithm used for data clustering, which groups unlabeled data points into groups or clusters (Kavlakoglu & Winland, 2025). K-Mean clustering assigns datapoints to the nearest randomly centroid. It then reissues the centroids based on the mean of the datapoints within the group. This process repeats until there no significant change. The elbow method will be used to determine the K value. Once we have our K value, we will generate clusters which will return metrics to determine the significance of each cluster. Expected outcome is all data points assigned to a cluster.
2. Summarize one assumption of the clustering technique.
 - One assumption of K-Means clustering is clusters are assumed to be spherical, which means their radius is approximately equal in all directions. The cluster center is assigned to the mean that the algorithm determines from the average of the data points within a cluster. Because of this presumption, K-means is susceptible to non-spherical or elongated clusters(GeeksforGeeks, 2025). Another assumption is that the scale of the variables can influence clustering results. To avoid needing to scale the variables used, we will be using the survey results(Item1 thru 8) as they are all already on the same scale.
3. List the packages or libraries you have chosen for Python or R, and justify how each item on the list supports the analysis.
 - In this analysis, we will be using Python within a Jupyter notebook. This gives us the ease and flexibility to have code and markdown cells in one easy to follow document. Python combined with Jupyter make it easy for not only implementing and reading code, but also creating the visual representations of the code all in one place making it easier to understand the data that is being worked with. Packages that will be used within Python are as follows:
 - Pandas and Numpy for preparation and analysis
 - Seaborn and Matplotlib for data visualization

- Scikit-learn for machine learning and predictive modeling

Part III: Data Preparation

C. Perform data preparation for the chosen data set by doing the following:

1. Describe one data preprocessing goal relevant to the clustering technique from part A1.
 - The goal is to have a clean data set that can be used for K-means clustering. Another goal will be to reduce the size of the dataset. The original dataset has more columns than what is needed for this analysis. There will be a more efficient approach and less cluttered.
2. Identify the initial data set variables you will use to perform the analysis for the clustering question from part A1, and label each as continuous or categorical.

- Item1 continuous
- Item2 continuous
- Item3 continuous
- Item4 continuous
- Item5 continuous
- Item6 continuous
- Item7 continuous
- Item8 continuous

These will also be renamed as part of the cleaning process to reflect what the survey items actually are. Below shows the new names as well as the original. It is also important to note we will not need to scale these variables as they are all on the same scale already. All variables listed below will still be continuous.

- "Item1": "Timely_response" (continuous)
- "Item2": "Timely_fixes" (continuous)
- "Item3": "Timely_replacements" (continuous)
- "Item4": "Reliability" (continuous)
- "Item5": "Options" (continuous)
- "Item6": "Respectful_response" (continuous)
- "Item7": "Courteous_exchange" (continuous)
- "Item8": "Active_listening" (continuous)

3. Explain each of the steps used to prepare the data for the analysis. Identify the code segment for each step.
 - needed packages will be imported into Python
 - `pd.read_csv` will be used to load data frame
 - `.info()` will be used to profile data frame
 - inspect data frame for duplicate values

- `.isnull().sum` will be used to identify possible missing values
 - if missing values exist, they will be treated with imputation
 - rename columns
 - create dataframe for variables
 - export cleaned dataframe
4. Provide a copy of the cleaned data set.
- Code for C3 & C4 is below.

```
In [67]: # code for c3
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats
import sklearn
import warnings
warnings.filterwarnings('ignore', category=FutureWarning)

# code for data frame
churn = pd.read_csv("C:/users/jjord/Documents/WGU/D212/d212_pa1/churn_clean.csv")

# Profile of Data Frame
churn.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 50 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   CaseOrder             10000 non-null  int64
 1   Customer_id           10000 non-null  object
 2   Interaction            10000 non-null  object
 3   UID                   10000 non-null  object
 4   City                  10000 non-null  object
 5   State                 10000 non-null  object
 6   County                10000 non-null  object
 7   Zip                   10000 non-null  int64
 8   Lat                   10000 non-null  float64
 9   Lng                   10000 non-null  float64
10   Population            10000 non-null  int64
11   Area                  10000 non-null  object
12   TimeZone              10000 non-null  object
13   Job                   10000 non-null  object
14   Children              10000 non-null  int64
15   Age                   10000 non-null  int64
16   Income                10000 non-null  float64
17   Marital               10000 non-null  object
18   Gender                10000 non-null  object
19   Churn                 10000 non-null  object
20   Outage_sec_perweek    10000 non-null  float64
21   Email                 10000 non-null  int64
22   Contacts              10000 non-null  int64
23   Yearly_equip_failure  10000 non-null  int64
24   Techie                10000 non-null  object
25   Contract              10000 non-null  object
26   Port_modem            10000 non-null  object
27   Tablet                10000 non-null  object
28   InternetService       7871 non-null   object
29   Phone                 10000 non-null  object
30   Multiple              10000 non-null  object
31   OnlineSecurity        10000 non-null  object
32   OnlineBackup          10000 non-null  object
33   DeviceProtection      10000 non-null  object
34   TechSupport           10000 non-null  object
35   StreamingTV           10000 non-null  object
36   StreamingMovies       10000 non-null  object
37   PaperlessBilling      10000 non-null  object
38   PaymentMethod         10000 non-null  object
39   Tenure                10000 non-null  float64
40   MonthlyCharge         10000 non-null  float64
41   Bandwidth_GB_Year     10000 non-null  float64
42   Item1                 10000 non-null  int64
43   Item2                 10000 non-null  int64
44   Item3                 10000 non-null  int64
45   Item4                 10000 non-null  int64
46   Item5                 10000 non-null  int64
47   Item6                 10000 non-null  int64
48   Item7                 10000 non-null  int64
49   Item8                 10000 non-null  int64

```

dtypes: float64(7), int64(16), object(27)

memory usage: 3.8+ MB

```
In [68]: # Missing values using isnull  
churn.isnull().sum()
```

```

Out[68]: CaseOrder      0
         Customer_id    0
         Interaction     0
         UID            0
         City           0
         State          0
         County         0
         Zip            0
         Lat            0
         Lng            0
         Population     0
         Area           0
         TimeZone       0
         Job            0
         Children       0
         Age            0
         Income         0
         Marital        0
         Gender         0
         Churn          0
         Outage_sec_perweek 0
         Email          0
         Contacts       0
         Yearly_equip_failure 0
         Techie         0
         Contract       0
         Port_modem     0
         Tablet         0
         InternetService 2129
         Phone          0
         Multiple       0
         OnlineSecurity 0
         OnlineBackup   0
         DeviceProtection 0
         TechSupport    0
         StreamingTV    0
         StreamingMovies 0
         PaperlessBilling 0
         PaymentMethod  0
         Tenure         0
         MonthlyCharge  0
         Bandwidth_GB_Year 0
         Item1          0
         Item2          0
         Item3          0
         Item4          0
         Item5          0
         Item6          0
         Item7          0
         Item8          0
         dtype: int64

```

```

In [69]: # Look at unique values for InternetService
         churn.InternetService.unique()

```

```
Out[69]: array(['Fiber Optic', 'DSL', nan], dtype=object)
```

```
In [70]: # Impute nulls for InternetService
churn["InternetService"].fillna("No Internet Service", inplace=True)
```

```
In [71]: # verify InternetService no longer has nulls
churn["InternetService"].isnull().sum()
```

```
Out[71]: 0
```

Columns Zip, Lat, Lng currently are floats or ints. There are not quantitative so will change datatype to object. These values are not meant for any type of calculation. Since the rest of the data types for categorical are set to object, these will also.

```
In [72]: obj_columns = ["Zip", "Lat", "Lng"]
churn[obj_columns] = churn[obj_columns].astype(object)
```

```
In [73]: # rename item columns
churn.rename(
    columns={
        "Item1": "Timely_response",
        "Item2": "Timely_fixes",
        "Item3": "Timely_replacements",
        "Item4": "Reliability",
        "Item5": "Options",
        "Item6": "Respectful_response",
        "Item7": "Courteous_exchange",
        "Item8": "Active_listening",
    },
    inplace=True,
)
```

```
In [74]: #profile df after clean
churn.info()
```



```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 50 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   CaseOrder                            10000 non-null  int64
 1   Customer_id                          10000 non-null  object
 2   Interaction                           10000 non-null  object
 3   UID                                  10000 non-null  object
 4   City                                 10000 non-null  object
 5   State                                10000 non-null  object
 6   County                               10000 non-null  object
 7   Zip                                  10000 non-null  object
 8   Lat                                  10000 non-null  object
 9   Lng                                  10000 non-null  object
10   Population                           10000 non-null  int64
11   Area                                 10000 non-null  object
12   TimeZone                             10000 non-null  object
13   Job                                  10000 non-null  object
14   Children                             10000 non-null  int64
15   Age                                  10000 non-null  int64
16   Income                               10000 non-null  float64
17   Marital                              10000 non-null  object
18   Gender                               10000 non-null  object
19   Churn                                10000 non-null  object
20   Outage_sec_perweek                   10000 non-null  float64
21   Email                                10000 non-null  int64
22   Contacts                             10000 non-null  int64
23   Yearly_equip_failure                  10000 non-null  int64
24   Techie                               10000 non-null  object
25   Contract                             10000 non-null  object
26   Port_modem                           10000 non-null  object
27   Tablet                               10000 non-null  object
28   InternetService                      10000 non-null  object
29   Phone                                10000 non-null  object
30   Multiple                             10000 non-null  object
31   OnlineSecurity                       10000 non-null  object
32   OnlineBackup                         10000 non-null  object
33   DeviceProtection                     10000 non-null  object
34   TechSupport                          10000 non-null  object
35   StreamingTV                          10000 non-null  object
36   StreamingMovies                      10000 non-null  object
37   PaperlessBilling                     10000 non-null  object
38   PaymentMethod                        10000 non-null  object
39   Tenure                               10000 non-null  float64
40   MonthlyCharge                        10000 non-null  float64
41   Bandwidth_GB_Year                    10000 non-null  float64
42   Timely_response                       10000 non-null  int64
43   Timely_fixes                          10000 non-null  int64
44   Timely_replacements                   10000 non-null  int64
45   Reliability                          10000 non-null  int64
46   Options                              10000 non-null  int64
47   Respectful_response                   10000 non-null  int64
48   Courteous_exchange                   10000 non-null  int64
49   Active_listening                      10000 non-null  int64

```

dtypes: float64(5), int64(15), object(30)

memory usage: 3.8+ MB

```
In [75]: # code for outliers.
        churn.describe()
```

```
Out[75]:
```

	CaseOrder	Population	Children	Age	Income	Outage_sec_per
count	10000.00000	10000.000000	10000.0000	10000.000000	10000.000000	10000.000000
mean	5000.50000	9756.562400	2.0877	53.078400	39806.926771	10.000000
std	2886.89568	14432.698671	2.1472	20.698882	28199.916702	2.000000
min	1.00000	0.000000	0.0000	18.000000	348.670000	0.000000
25%	2500.75000	738.000000	0.0000	35.000000	19224.717500	8.000000
50%	5000.50000	2910.500000	1.0000	53.000000	33170.605000	10.000000
75%	7500.25000	13168.000000	3.0000	71.000000	53246.170000	11.000000
max	10000.00000	111850.000000	10.0000	89.000000	258900.700000	21.000000

```
In [76]: # dataframe for variables

        prep_churn = churn[["Timely_response", "Timely_fixes", "Timely_replacements", "Reliability"]]
```

```
In [77]: # profile df
        prep_churn.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Timely_response        10000 non-null  int64
1   Timely_fixes           10000 non-null  int64
2   Timely_replacements    10000 non-null  int64
3   Reliability            10000 non-null  int64
4   Options                10000 non-null  int64
5   Respectful_response    10000 non-null  int64
6   Courteous_exchange     10000 non-null  int64
7   Active_listening       10000 non-null  int64
dtypes: int64(8)
memory usage: 625.1 KB
```

```
In [78]: prep_churn.to_csv('prep_churn.csv', index=False)
```

Part IV: Analysis

D. Perform the data analysis and report on the results by doing the following:

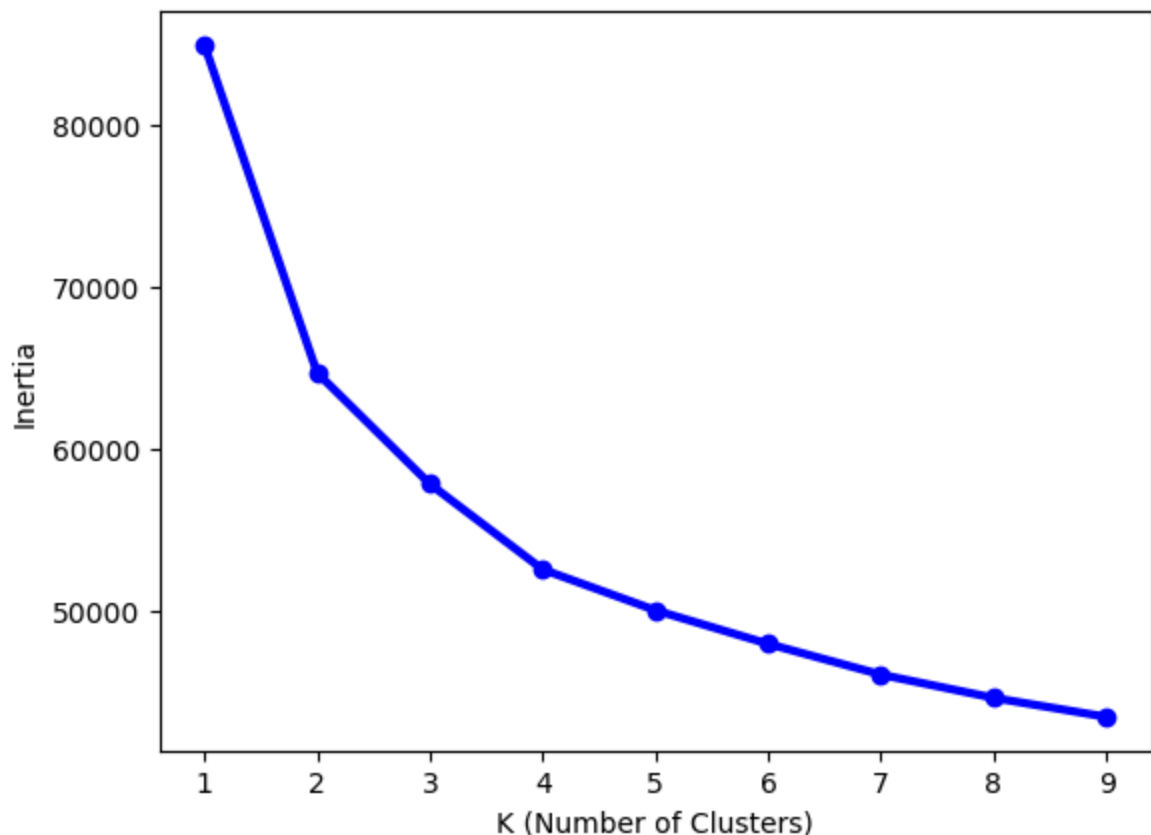
1. Determine the optimal number of clusters in the data set, and describe the method used to determine this number. By using the elbow method in the code below, it was determined that the optimal number of clusters is 2. The elbow method plots the inertia against different values of k. The optimal k is found at the elbow or bend where the curve begins to flatten, thus indicating diminishing returns. Again, the optimal number of clusters that will be used is 2(k=2).
2. Provide the code used to perform the clustering analysis technique.

Code for both D1 and D2 are below.

```
In [79]: # d1 code
from sklearn.cluster import KMeans
# number of clusters, or k, through inertia

inertia = []
for i in range(1,10):
    kmeanModel = KMeans(n_clusters=i, n_init=10)
    kmeanModel.fit(prepare_churn)
    inertia.append(kmeanModel.inertia_)

# elbow graph plot
plt.plot(range(1,10), inertia, 'o-', linewidth=3, color='blue')
plt.xlabel('K (Number of Clusters)')
plt.ylabel('Inertia')
plt.show()
```



In [80]: *# d2 code cont.*

kmean cluster creation

```
model = KMeans(n_clusters=2)
model.fit(prepare_churn)
model.labels_
```

Out[80]: array([1, 0, 0, ..., 1, 1, 0])

In [81]: *# d2 code cont.*

model.cluster_centers_

Out[81]: array([[2.83981154, 2.89163722, 2.92559874, 3.29642717, 3.74047899,
2.93678838, 3.02021987, 3.07263447],
[4.16673461, 4.14207093, 4.06991439, 3.70627803, 3.23583367,
4.07929066, 4.01752956, 3.93477375]])

In [82]: *# d2 code cont.*

```
from sklearn.metrics import silhouette_score
from sklearn.metrics import davies_bouldin_score
```

```
clusters_range = range(2, 10)
```

silhouette scores list

```
silhouette_scores = []
```

loop for clusters

```
for i in clusters_range:
    kmeanModel = KMeans(n_clusters=i, n_init=10)
    kmeanModel.fit(prepare_churn)
```

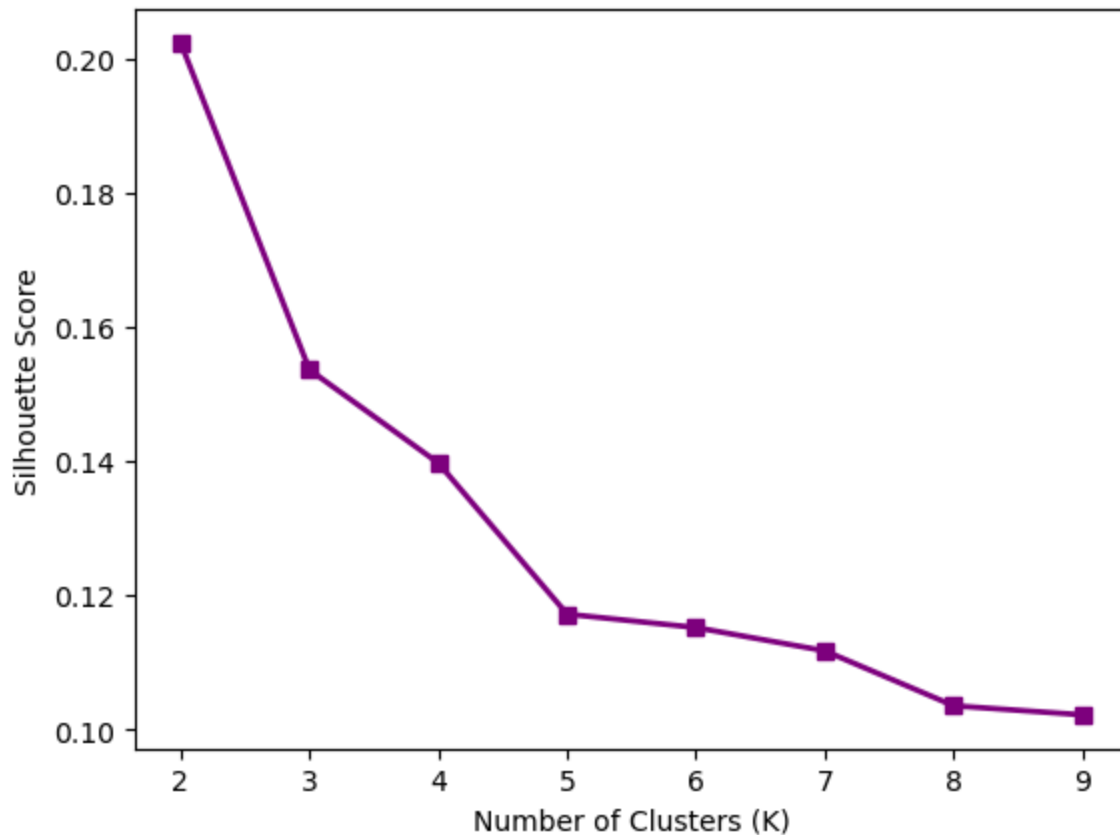
Calculate silhouette score

```
silhouette_avg = silhouette_score(prepare_churn, kmeanModel.labels_)
silhouette_scores.append(silhouette_avg)
print(f"cluster # = {i}, silhouette score is {silhouette_avg}")
```

visualizing the silhouette score

```
plt.plot(clusters_range, silhouette_scores, 's-', linewidth=2, color='purple')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Silhouette Score')
plt.show()
```

```
cluster # = 2, silhouette score is 0.2022991197361273
cluster # = 3, silhouette score is 0.15358953820977406
cluster # = 4, silhouette score is 0.13969378946244754
cluster # = 5, silhouette score is 0.11717812963651589
cluster # = 6, silhouette score is 0.11517361417045172
cluster # = 7, silhouette score is 0.11170352055876354
cluster # = 8, silhouette score is 0.10352313190859941
cluster # = 9, silhouette score is 0.10216778126618307
```



```
In [83]: # d2 code cont.
# WCSS/Inertia scoring
inertia = model.inertia_

# Labels
labels = model.labels_

# Silhouette scoring
silhouette_avg = silhouette_score(prepare_churn, labels)

# Davies-Bouldin Index
db_index = davies_bouldin_score(prepare_churn, labels)

# print metrics
print("WCSS (Inertia):", inertia)
print("Silhouette Score:", silhouette_avg)
print("Davies Bouldin Index:", db_index)
```

```
WCSS (Inertia): 64659.527968146045
Silhouette Score: 0.20229678323339656
Davies Bouldin Index: 1.7212071598822394
```

Part V: Data Summary and Implications

E. Summarize your data analysis by doing the following:

1. Explain the quality of the clusters created.

- WCSS(Inertia)

The WCSS or Inertia measures the compactness and variance of the clusters by calculating the sum of the squared distances between every data point and its assigned cluster centroid. The lower the score, the more compact the cluster is. The model has a WCSS score of 64659.55, indicating the datapoints within the clusters are moderately spreadout from the cluster centroids. This result is not ideal.

- Silhouette Score

The Silhouette Score is a metric that evaluates the quality of the clusters by measuring how close each point in one cluster is to points in neighboring clusters. Unlike the WCSS, a higher Silhouette score indicates the datapoint is more compact within its own cluster and further away from neighboring clusters. The range of the Silhouette score is 1 to -1. The model has a Silhouette score of .202 which is only a moderate level. This result is middle of the road as we would prefer to see it much closer to 1.

- Davies Bouldin Index

The Davies Bouldin Index is a metric that evaluates how well a clustering algorithm has performed, in this case, K-Means. It measures the average similarity between each cluster and its most similar one, calculated as the ratio of how spread out a cluster is compared to how far apart clusters are from each other(GeeksforGeeks, Davies-Bouldin index 2025). The best possible score for Davies Bouldin Index is 0, so a low score means better clustering. The model has a Davies Bouldin Index of 1.721, which is not horrible but also not ideal. This result we would prefer to be closer to 0.

2. Discuss the results and implications of your clustering analysis.

The elbow method was used to find the optimal K value for the model which turned out to be a k value of 2. This provided a Silhouette score of .202.

The results of the arrays represent the centroid of the cluster in an 8 feature space. An array for the 2 centroids was returned.

The center of the clusters from the array in the first row are [4.16642901, 4.1415767 , 4.06926054, 3.70666123, 3.23548584, 4.0790385 , 4.01690772, 3.9346099]. The center of the clusters from the array in the second row are [2.8393243 , 2.89137694, 2.9255549 , 3.29581615, 3.74111177, 2.93635828, 3.02023178, 3.07228442]. We are working with a 8 feature space, each row from the array has 8 values to represent the centroid of each cluster. Analyzing the results of the clusters, cluster 1 has higher values overall versus cluster 2. This lets us assume that cluster 2 has lower feature values in all 8 feature spaces, separating the datapoints within the cluster based on the values.

3. Discuss one limitation of your data analysis.

One of the limitations of K-Means Clustering is that it only uses continuous variables. That makes it difficult to get proper insight of the dataset when you are limited to a

certain variable type. Being able to use categorical variables would give more insight to the data within the dataset.

4. Recommend a course of action for the real-world organizational situation from part A1 based on the results and implications discussed in part E2.

It is recommended that further analysis be done using different continuous variables or even the other suggested clustering method due to the mediocrity of the results. Spending more time on the analysis will give better insight, allowing for a more informed decision than what could be made now. After further analysis has been done, the company can then make an informed decision on the research question.

Part VI: Demonstration

F. Provide a Panopto video recording that includes the presenter and a vocalized demonstration of the functionality of the code used for the analysis of the programming environment.

- <https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=ddb6e067-bcc1-4e5f-9e40-b469016b99d7>

G. List the web sources used to acquire data or segments of third-party code to support the application. Ensure the web sources are reliable.

None used.

H. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

WGU. (n.d.). Churn Data Consideration and Dictionary. Data Files and Associated Dictionary Files.
<https://access.wgu.edu/ASP3/aap/content/d9rkejv84kd9rk30fi2l.zip>

Kavlakoglu, E., & Winland, V. (2025, November 17). What is K-means clustering?. IBM. <https://www.ibm.com/think/topics/k-means-clustering>

GeeksforGeeks. (2025, July 23). Demonstration of K-means assumptions. <https://www.geeksforgeeks.org/machine-learning/demonstration-of-k-means-assumptions/>

GeeksforGeeks (Ed.). (2025a, July 23). Davies-Bouldin index.

GeeksforGeeks. <https://www.geeksforgeeks.org/machine-learning/davies-bouldin-index/>